The abrupt–strongly eroded dichotomy

The main ideas of the proof

Jorge González Cázares[1]

David Kramer-Bang[2]

Aleksandar Mijatović[3]

[1]Department of Probability and Statistics, IIMAS–UNAM, Mexico

[2]Department of Mathematics, Aarhus University, Denmark

[3]Department of Statistics, University of Warwick
& The Alan Turing Institute, United Kingdom







Finiteness on one interval is enough

Let X be a real Lévy process of infinite variation. Define

$$J_X(I) = \int_0^1 \mathbb{P}(X_t/t \in I)\frac{\mathrm{d}t}{t}.$$

Throughout, intervals are nonempty, bounded and open.

Dichotomy
Either $J_X(I) < \infty$ for every interval I (abrupt), or $J_X(I) = \infty$ for every interval I (strongly eroded).

Abrupt: steep local minima; strongly eroded: a C^1 convex minorant.

Proof target:

$$J_X(I) < \infty \text{ for one } I \quad \Longrightarrow \quad J_X(J) < \infty \text{ for every } J.$$

Characterisations: Vigon, *Stochastic Process. Appl.* 103 (2003), 155–168.

González Cázares, Kramer-Bang and Mijatović, *AIHP* 60 (2024), 2352–2381.

The interval-integrated Fourier kernel

Use $\mathbb{E}e^{iuX_t} = e^{t\psi(u)}$ and, for $u > 0$, write

$$Q(u) = -\Re\psi(u), \qquad a(u) = \frac{1+Q(u)}{u}, \qquad b(u) = \frac{\Im\psi(u)}{u}.$$

$$\Phi_I(u) = \int_I \frac{a(u)}{a(u)^2 + (r-b(u))^2}\,\mathrm{d}r, \qquad 0 < \Phi_I(u) \le \pi.$$

$$\begin{aligned}\int_0^\infty e^{-t}\mathbb{P}(X_t/t \in I)\frac{\mathrm{d}t}{t} &= \frac{1}{2\pi}\int_{\mathbb{R}}\mathrm{d}u\int_I \Re\frac{1}{1+iur-\psi(u)}\,\mathrm{d}r \\ &= \frac{1}{\pi}\int_0^\infty \Phi_I(u)\frac{\mathrm{d}u}{u}.\end{aligned}$$

Thus $J_X(I) < \infty$ iff $\int_1^\infty \Phi_I(u)\,\mathrm{d}u/u < \infty$.

Low frequencies are harmless, since

$$\frac{\Phi_I(u)}{u} \le \frac{|I|}{u\,a(u)} = \frac{|I|}{1+Q(u)} \le |I|.$$

Fourier identity: Theorem B.2 of González Cázares, Kramer-Bang and Mijatović, *AIHP* 60 (2024), 2352–2381.

Remove large jumps

Delete jumps with $|x| \geq 1$, of finite intensity $\lambda = \nu(\{|x| \geq 1\})$.

$$\left|\mathbb{P}(X_t/t \in I) - \mathbb{P}(\widetilde{X}_t/t \in I)\right| \leq \lambda t.$$

- Finiteness of $J_X(I)$ is unchanged.
- Infinite variation is preserved; the Gaussian part is retained.

We may therefore use

$$\psi(u) = i\gamma u - \tfrac{1}{2}\sigma^2 u^2 + \int_{(-1,1)} (e^{iux} - 1 - iux)\nu(\,\mathrm{d}x).$$

Smoothen the oscillating characteristics

$$H(u) = \sigma^2 u + \int_{(-1,1)} \frac{ux^2}{1+u^2x^2}\nu(\,\mathrm{d}x),$$
$$D(u) = \gamma - \int_{(-1,1)} \frac{u^2x^3}{1+u^2x^2}\nu(\,\mathrm{d}x).$$

Uniform estimates, with universal $c_0, C_0 > 0$:

$$|uH'(u)|,\ |uD'(u)| \le H(u),$$
$$\sup_{|z|\le 1} |\psi(uz) - iuzD(u)| \le C_0 uH(u),$$
$$\int_{-1}^{1} Q(uz)\,\mathrm{d}z, \quad \int_{1}^{2} Q(uz)\,\mathrm{d}z \ge c_0 uH(u).$$

Infinite variation gives $\displaystyle\int_1^\infty H(u)\frac{\mathrm{d}u}{u} = \infty.$

Finite occupation gives a finite energy integral

Fix I with $\int_1^\infty \Phi_I(u)\,\mathrm{d}u/u < \infty$.

On $u \le v \le 2u$, the smoothed estimates imply

$$a(v)^2 + (r - b(v))^2 \le C_I\big(1 + H(u)^2 + D(u)^2\big), \qquad r \in I.$$

Using $a(v) \ge Q(v)/v$ and the averaged lower bound for Q,

$$\int_u^{2u} \Phi_I(v)\frac{\mathrm{d}v}{v} \ge c_I \frac{H(u)}{1 + H(u)^2 + D(u)^2}.$$

Integrate in $\mathrm{d}u/u$ and apply Tonelli:

$$\int_1^\infty \frac{H(u)}{1 + H(u)^2 + D(u)^2}\frac{\mathrm{d}u}{u} < \infty. \tag{1}$$

The running maximum escapes every linear bound

Set $h(s) = H(e^s)$, $d(s) = D(e^s)$ and $R(s) = \sqrt{h(s)^2 + d(s)^2}$.

$$\int_0^\infty \frac{h(s)}{1 + R(s)^2}\,\mathrm{d}s < \infty \qquad \text{by (1), with } u = e^s.$$

$$\left|\frac{\mathrm{d}}{\mathrm{d}s}\arctan R(s)\right| \le \frac{\sqrt{2}\,h(s)}{1 + R(s)^2}.$$

- Thus $(\arctan R)' \in L^1(0,\infty)$, so $\arctan R$ converges.
- A limit below $\pi/2$ would force $\int_0^\infty h(s)\,\mathrm{d}s = \int_1^\infty H(u)\frac{\mathrm{d}u}{u} < \infty$, contradicting infinite variation and implying $R(s) \to \infty$.

Hence $R(\log u) \le H(u) + |D(u)| \to \infty$. The smoothed estimates give

$$P(u) := \sup_{0 \le v \le u} |\psi(v)| \ge c\,u\big(H(u) + |D(u)|\big).$$

$$P(u)/u \longrightarrow \infty.$$

The key local averaging lemma

For fixed $M > 0$, consider the exceptional frequencies

$$E_M = \{u \geq 1 : |\psi(u)| \leq Mu\}.$$

For every sufficiently large $u \in E_M$, there is a centred interval $J_u = (u - \ell_u, u + \ell_u)$, with $0 < \ell_u \leq u/10$, such that

$$\frac{1}{|J_u|} \int_{J_u} \Phi_I(w)\,\mathrm{d}w \geq c_{I,M} > 0.$$

A small characteristic exponent forces nearby occupation mass.

We define an appropriate constant $\varepsilon > 0$ in terms of $I = (a, b)$.
If $Q(u) \geq \varepsilon u$, the kernel is already bounded below at u:
$\varepsilon \leq a(u) \leq M + 1$ and hence $\Phi_I(u) \geq \frac{|I|\varepsilon}{(M+1)^2+(|a|+|b|+M)^2}$.
Continuity yields $\ell_u > 0$.
We focus on $Q(u) < \varepsilon u$.

Translate a controlled piece of the exponent

Lévy–Khintchine and Cauchy–Schwarz give

$$|\psi(u+v)-\psi(u)-\psi(v)| \le 2\sqrt{Q(u)Q(v)}.$$

Choose K large relative to I, M, and the first exit radius

$$\rho = \inf\{v > 0 : |\psi(v)| = Ku\} = o(u).$$

Set $\chi(z) = \psi(\rho z)/u$ and $q(z) = -\Re\chi(z)$ for $|z| \le 1$.

$$|\chi(z)| \le K, \qquad |\chi(1)| = K, \qquad \frac{\psi(u+\rho z)}{u} = \frac{\psi(u)}{u} + \chi(z) + e(z),$$

$$|e(z)| \le 2\sqrt{\varepsilon q(z)}, \qquad T = \frac{\rho H(\rho)}{u}, \quad d = \frac{\rho D(\rho)}{u}.$$

Two possibilities: appreciable T, or an almost linear χ.

Case 1: enough real part nearby

Suppose $T \geq \delta > 0$. The smoothed estimates give

$$0 \leq q(z) \leq K, \qquad \int_{-1}^{1} q(z)\,\mathrm{d}z \geq c_0 T.$$

Thus $V := \{z \in [-1,1] : q(z) \geq c_0\delta/4\}$ has uniformly positive length.

The increment estimate also yields

$$\frac{Q(u+\rho z)}{u} \geq \left(\sqrt{q(z)} - \sqrt{Q(u)/u}\right)^2.$$

Choose ε small relative to δ. On V,

$$a(u+\rho z) \geq c > 0, \qquad a(u+\rho z) + |b(u+\rho z)| \leq C.$$

The kernel $\Phi_I(u+\rho z)$ is uniformly positive on V, so its average on $(u-\rho, u+\rho)$ is positive.

Case 2: an almost linear imaginary part

Suppose $T < \delta$. Then

$$\sup_{|z|\le 1} |\chi(z) - idz| \le C_0\delta, \qquad \big||d| - K\big| \le C_0\delta.$$

Write $I = (\alpha, \beta)$, $L = \beta - \alpha$ and $b_0 = \Im\psi(u)/u$. The line $b_0 + dz$ crosses the middle half of I on a set

$$V = \{z : b_0 + dz \in [\alpha + L/4, \beta - L/4]\} \subset [-1, 1], \qquad |V| \ge L/(3K).$$

For small δ, ε and large u, on V,

$$|b(u + \rho z) - (b_0 + dz)| \le L/8, \qquad a(u + \rho z) \le L/8.$$

The kernel is narrow and its centre lies inside I: $\Phi_I(u + \rho z) \ge \pi/2$. Again the local average is positive.

Exceptional frequencies have finite logarithmic measure

For each large $u \in E_M$, the local lemma supplies J_u with

$$\int_{J_u} \Phi_I(w)\,\mathrm{d}w \geq c|J_u|, \qquad \text{radius}(J_u) \leq u/10.$$

Select disjoint intervals J_k whose fivefold dilations cover $E_M \cap [1, n]$ on a bounded truncation $[1, n]$. Then

$$\int_{5J_k} \frac{\mathrm{d}w}{w} \leq C\frac{|J_k|}{u_k} \leq C' \int_{J_k} \Phi_I(w)\frac{\mathrm{d}w}{w}.$$

Sum over disjoint J_k and let the truncation grow:

$$\int_{E_M} \frac{\mathrm{d}u}{u} \leq C'' \int_0^\infty \Phi_I(u)\frac{\mathrm{d}u}{u} + C''' < \infty.$$

Compare any two slope intervals

Fix J, take $I \cup J \subset [-R, R]$, and set $M = 2(R+1)$. For $u \geq 1$ outside E_M and $|r| \leq R$,

$$\tfrac{1}{2}|\psi(u)| \leq |1 + iur - \psi(u)| \leq \tfrac{3}{2}|\psi(u)|.$$

The numerator of the resolvent is independent of r:

$$\Re \frac{1}{1 + iur - \psi(u)} = \frac{1 + Q(u)}{|1 + iur - \psi(u)|^2}.$$

Hence $\Phi_J(u) \leq 9\frac{|J|}{|I|}\Phi_I(u)$ outside E_M.

$$\int_1^\infty \Phi_J(u)\frac{\mathrm{d}u}{u} \leq 9\frac{|J|}{|I|}\int_1^\infty \Phi_I(u)\frac{\mathrm{d}u}{u} + \pi\int_{E_M}\frac{\mathrm{d}u}{u} < \infty.$$

The Fourier identity gives $J_X(J) < \infty$ for every J.

Conclusion: what the proof controls

Finite logarithmic occupation on one interval forces finiteness on all bounded intervals. Every infinite-variation Lévy process is therefore abrupt or strongly eroded.

- Escape: smoothed characteristics force $P(u)/u \to \infty$.
- Averaging: the remaining troughs have finite logarithmic measure.
- Comparison: interval-integrated resolvents share finiteness.

The proof controls integrals over slopes; exceptional drifts with polar points remain possible.